

Vendor Accreditation System – Requirements

1. Introduction

1.1 Purpose

This document details the requirements for a Vendor Accreditation System that streamlines event accreditation by automating vendor form submission, ID card generation with QR codes, and access validation via a mobile guard/checker application (Android and iOS). The system aims to facilitate seamless interaction between four user roles—Admin, Vendor, Printer, and Guard—ensuring efficient management of accreditation processes, secure access control, and real-time tracking.

1.2 Scope

The system will consist of:

- A **web portal** with distinct dashboards for Admin, Vendor, and Printer.
- A **mobile application** for Guards to scan QR codes and verify access rights.
- Integration of a **QR Code generation and mapping system** to uniquely identify printed ID cards.
- Role-based access and permission management.
- Notification and reporting modules to track and log system activities.

1.3 Definitions & Acronyms

- **Admin:** The event organizer or system owner managing vendors, printers, and guards.
- **Vendor:** The party submitting forms and assigning access levels (e.g., dinner, VIP).

- **Printer:** The service provider printing ID cards with embedded QR codes.
 - **Guard/Checker:** The mobile app user responsible for scanning QR codes at the event.
 - **QR Code:** A machine-readable optical label containing a unique identifier mapped to each printed ID card.
 - **API:** Application Programming Interface.
 - **UI/UX:** User Interface/User Experience.
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2. System Overview

The system is designed as a multi-tier solution consisting of:

- **Frontend:** Web-based interfaces for Admin, Vendor, and Printer roles; Mobile app interface for Guards.
 - **Backend:** Server-side logic handling authentication, form processing, QR mapping, API integrations (including with AI if required), and notifications.
 - **Database:** A relational or NoSQL database for storing user data, form details, QR mapping, and access logs.
 - **External Interfaces:** Integration with third-party APIs (e.g., Claude AI) for potential automated suggestions or data processing.
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3. Functional Requirements

3.1 User Authentication & Authorization

- **Login/Logout:** Secure authentication for all users using email and password.
- **Role-Based Access Control:**
 - **Admin:** Full system management rights.
 - **Vendor:** Access to form submission, tracking, and access level assignments.
 - **Printer:** Access to approved forms, printing functionalities, and QR code mapping.

- **Guard:** Access via mobile app to scan and validate QR codes.
- **Password Recovery:** "Forgot Password" and reset functionality.

3.2 Admin Functions

- **Dashboard:** Overview of vendor submissions, form statuses, printer job queue, and guard activity logs.
- **User Management:**
 - Add, update, or remove vendors and printers.
 - Grant mobile app access to guards.
- **Form & ID Card Management:**
 - Review submitted forms.
 - Approve or reject forms.
 - Assign approved forms to printers.
- **QR Code Mapping:**
 - After a form is approved and an ID card printed, map the printed ID card's unique identifier with the corresponding QR code.
- **Notifications & Reporting:**
 - Email and in-app notifications for form status changes.
 - Generate reports on vendor submissions, printing status, and guard scan logs.

3.3 Vendor Functions

- **Dashboard:** View the number of allocated forms, current submission status, and any approval feedback.
- **Form Submission:**
 - Fill out forms with required details and assign access levels (e.g., dinner, VIP, backstage).
 - Submit forms for admin approval.

- **Status Tracking:** Monitor the approval status and receive notifications on changes.
- **Feedback Loop:** If a form is rejected, view detailed feedback for resubmission.

3.4 Printer Functions

- **Dashboard:**
 - Access to a list of approved forms ready for printing.
 - View assigned printing tasks and job statuses.
- **Printing Process:**
 - Print ID cards that include a unique QR code.
 - After printing, scan or confirm the QR code mapping to the unique ID of the card.
- **Status Update:** Mark printed jobs as "Completed" to update system records.

3.5 Guard/Checker Functions (Mobile App)

- **User Authentication:** Secure login for guard users.
- **QR Code Scanning:**
 - Use the mobile app's camera to scan QR codes on ID cards.
 - Retrieve and display associated user details such as name, photo, and access level.
- **Access Verification:**
 - Validate access permissions (e.g., dinner, VIP) as defined by the vendor.
 - Provide immediate feedback (allow/deny) based on validation.
- **Logging:**
 - Record each scan event along with timestamp and validation outcome.
 - Enable report generation and activity log review.

3.6 Notification & Reporting

- **Email/SMS Alerts:**
 - Notify vendors when forms are approved or rejected.
 - Alert printers when new jobs are assigned.
 - Inform guards of any changes in access levels.
 - **Activity Reports:**
 - Generate logs on form submissions, printing status, and guard scans.
 - Export reports in common formats (e.g., CSV, PDF).
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4. Non-Functional Requirements

4.1 Performance

- **Response Time:** All actions (form submissions, QR scans) must be processed within 2–3 seconds.
- **Scalability:** The system must handle a high volume of concurrent users during peak event periods.

4.2 Security

- **Data Encryption:** Use HTTPS for all data transmissions; encrypt sensitive data (e.g., passwords) at rest.
- **Access Control:** Ensure strict role-based access to avoid unauthorized operations.
- **Input Validation:** Sanitize and validate all user inputs to prevent injection attacks.
- **Audit Logging:** Maintain logs for all critical actions for future audits.

4.3 Usability

- **User Interface:** Clean and intuitive design optimized for both desktop and mobile environments.
- **Accessibility:** Follow accessibility guidelines (e.g., WCAG) to ensure usability for all users.

- **Localization:** Ability to support multiple languages if required.

4.4 Reliability & Availability

- **Uptime:** Aim for 99.5% uptime, especially during event periods.
- **Backup & Recovery:** Implement regular data backups and a disaster recovery plan.

4.5 Maintainability

- **Modular Design:** Develop the system in modular components for ease of updates and feature additions.
 - **Documentation:** Provide comprehensive technical and user documentation.
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5. System Architecture

5.1 Overview

- **Frontend:**
 - **Web Portal:** Built using modern JavaScript frameworks (React, Angular, or Vue.js) ensuring responsive design.
 - **Mobile App:** Cross-platform development (using React Native, Flutter, or native development for Android and iOS) for the guard/checker application.
- **Backend:**
 - API-driven backend developed using Node.js (Express) or Python (Flask/Django).
 - Handles user authentication, form processing, QR mapping, and notifications.
- **Database:**
 - Use a relational database (e.g., PostgreSQL, MySQL) for structured data and transactions.
 - Alternatively, a NoSQL database (e.g., MongoDB) for flexibility if required.

- **Integration:**

- Integration with third-party services (Claude AI API) if needed for additional functionality.
- Email/SMS service for notifications.

5.2 Data Flow

1. **User Interaction:** Users (Admin, Vendor, Printer, Guard) interact through the web portal or mobile app.
 2. **Request Handling:** Frontend sends requests to the backend API for authentication, form submissions, or QR scan validation.
 3. **Database Operations:** The backend processes requests and interacts with the database to store or retrieve data.
 4. **QR Mapping:** Upon printing, the backend maps a generated QR code with the unique ID of the ID card.
 5. **Guard Verification:** The mobile app fetches the mapped details for validation when a QR code is scanned.
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6. User Flow Diagrams & Use Cases

6.1 Admin Use Cases

- **Manage Users:** Create, update, or remove vendors, printers, and guards.
- **Approve Forms:** Review vendor submissions and decide approval/rejection.
- **Assign Printing Jobs:** Allocate approved forms to printers.
- **QR Mapping:** Confirm the mapping of printed ID cards with their QR codes.
- **Reporting:** Generate detailed activity and status reports.

6.2 Vendor Use Cases

- **Submit Forms:** Fill out and submit accreditation forms with assigned access levels.
- **View Status:** Monitor the progress of submitted forms and view any feedback.

- **Resubmit Forms:** Update and resubmit forms if rejected.

6.3 Printer Use Cases

- **Access Print Queue:** View and process approved forms for printing.
- **Print ID Cards:** Print the ID cards containing unique QR codes.
- **Map QR Codes:** Scan and map printed QR codes to the corresponding ID card IDs.
- **Update Status:** Mark print jobs as completed.

6.4 Guard/Checker Use Cases

- **QR Code Scanning:** Scan ID card QR codes using the mobile app.
 - **Access Verification:** Display user details and validate access rights.
 - **Log Entry:** Record each scan event with outcome and timestamp.
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7. Timeline & Milestones

7.1 Overall Duration: 3 Months

- **Month 1:**
 - UI/UX design for Web Portal (Admin, Vendor, Printer) and Mobile App (Guard).
 - Approval of design mockups and interactive prototypes.
- **Month 2:**
 - Backend development including user authentication, form management, QR code generation/mapping, and API integrations.
 - Initial mobile app development for the guard application.
- **Month 3:**
 - Testing (unit, integration, and beta testing across all platforms).
 - Final refinements, security audits, and performance optimization.

- Deployment of the web portal and publication of the mobile app on Android and iOS.
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8. Assumptions & Constraints

8.1 Assumptions

- The client will provide timely feedback during the design and development phases.
- Third-party APIs (e.g., Claude AI) are available and functional within project timelines.
- Required hosting and infrastructure are set up by the client or provided as part of the project.

8.2 Constraints

- The system must be scalable to handle high volumes during peak event periods.
 - All user data must be secured according to industry best practices.
 - The mobile app must support both Android and iOS platforms with consistent functionality.
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9. Deliverables

- **Web Portal:**
 - Fully functional dashboards for Admin, Vendor, and Printer.
 - Secure authentication and role-based access.
 - End-to-end form submission, approval, printing, and QR mapping process.
- **Mobile App:**
 - Guard/Checker app for QR scanning and access verification (Android & iOS).
- **Backend:**

- Robust API supporting user management, form processing, and notifications.
 - Database schema and integration.
 - **Documentation:**
 - Technical documentation covering system architecture, API endpoints, and integration details.
 - User manuals for Admin, Vendor, Printer, and Guard roles.
 - **Testing & Deployment:**
 - Comprehensive testing reports.
 - Deployment on designated domains and mobile app stores.
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10. Future Enhancements

- Integration of advanced analytics for real-time reporting.
 - Automated escalation for form rejections or printer job failures.
 - Additional access control features (e.g., facial recognition, biometrics) for enhanced security.
 - Enhanced notification system via SMS and push notifications.
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This document captures all functional and non-functional requirements for the proposed Vendor Accreditation System with QR-based ID verification and mobile guard/checker app. It serves as the foundation for subsequent design, development, and testing phases.

Below is a comprehensive requirements document for the Vendor Accreditation System with QR-based ID verification and a mobile guard/checker application.

Vendor Accreditation System – Requirements Document

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